

TSLA on CMS DE-SynPUF: self-harm prediction from prior mental-health codes

Reproduction of Chen, Aseltine, Wang & Chen (JASA 2024) on public synthetic Medicare claims

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Note on data. DE-SynPUF is synthetic and deliberately perturbed. CMS describes it as having very limited inferential value. The results below show that the pipeline works end to end. They are not clinical findings. The paper’s original data (Kansas HIN EHR) is not public.

1. Study design

- **Data:** CMS DE-SynPUF sample 1 (2008–2010). Beneficiary files plus inpatient and outpatient claims (116,352 beneficiaries).
- **Outcome:** self-harm, defined as the first claim with ICD-9 E950–E959. That claim’s date is the index date.
- **Cases:** 196 people with a self-harm code and at least one earlier claim. Cases whose first visit was the self-harm visit were dropped, as in the paper.
- **Controls:** 1:10 nested case-control (risk-set) sampling, giving 1,960 controls. Controls match the case on sex and birth year, have no self-harm by the index date, have a non-self-harm claim within ± 30 days of it, and have prior claims history.
- **Analysis set:** 2,156 records (9.1% cases).
- **Features:** each ICD-9 mental-disorder code (290–319) recorded before the index date becomes a binary indicator. Codes present in fewer than 5 records were removed, leaving **101 codes**. Age (standardized) and sex are unpenalized covariates.
- **Hierarchy:** the codes form a five-level tree: full code (101 leaves), then 4-digit (56 nodes), 3-digit (22), section (4: 290–294, 295–299, 300–316, 317–319), and chapter (1).
- **Evaluation:** 3 repeated 70/30 train/test splits by matched set, with 649 test records and 59 test cases per split. Tuning uses 3-fold cross-validation on the training set with AUC as the loss.

2. Results by split

Split	Method	AUC	AUPRC	Sens @5%	PPV @5%	Sens @10%	PPV @10%
1	TSLA	0.594	0.129	0.068	0.121	0.203	0.185
1	Enet	0.588	0.130	0.068	0.121	0.203	0.185
1	Enet2	0.589	0.143	0.119	0.212	0.169	0.154
2	TSLA	0.496	0.088	0.034	0.061	0.068	0.062
2	Enet	0.541	0.123	0.119	0.212	0.203	0.185
2	Enet2	0.576	0.151	0.136	0.242	0.186	0.169
3	TSLA	0.638	0.177	0.153	0.273	0.203	0.185
3	Enet	0.535	0.139	0.136	0.242	0.186	0.169
3	Enet2	0.618	0.180	0.153	0.273	0.254	0.231

TSLA tuning per split (lambda.min / alpha.min): split 1: 12.7 / 0; split 2: 215.7 / 0; split 3: 11.7 / 0.
TSLA fit time was about 350 s per split.

3. Mean over 3 splits

Method	AUC	AUPRC	Sens @5%	PPV @5%	Sens @10%	PPV @10%
TSLA	0.576	0.131	0.085	0.152	0.158	0.144
Enet	0.555	0.131	0.107	0.192	0.198	0.179
Enet2	0.594	0.158	0.136	0.242	0.203	0.185

Enet is an elastic net ($\alpha = 0.5$) on the 101 full codes. *Enet2* is the same model on codes collapsed to 3 digits. Random guessing gives an AUPRC of about 0.091, the test case share. Sensitivity and PPV are measured when the top 5% or 10% of predicted risk scores are flagged.

Reading the results. All three methods perform only a little better than chance. *Enet2* has the highest mean AUC (0.594), TSLA is next (0.576), and *Enet* is lowest (0.555). The per-split variation is large: TSLA's AUC ranges from 0.496 to 0.638. With only 59 test cases per split, the differences between methods are within noise, so no method is clearly better. This is expected on synthetic data, where CMS scrambled much of the real correlation between diagnoses. The paper reported a clear advantage for TSLA on real EHR data.

4. Features selected by TSLA (full-data fit)

TSLA kept **34 features**. Most are individual codes, and a few are merged groups formed by OR-ing the codes under one tree node. Prevalence is the share of cases or controls with the feature. The odds ratio comes from an ordinary logistic refit on the kept features. It is a rough guide only: it is estimated after selection, has no valid p-values, and is unstable for rare codes.

Level	Node	Codes merged	Description	Cases	Controls	Refit OR
section	290-294	15 of 15	Senile dementia, uncomplicated + others	17.9%	15.7%	0.85
section	300-316	20 of 29	Anxiety state, unspecified + others	33.2%	23.1%	1.12
section	317-319	3 of 3	Mild intellectual disabilities; Severe intellectual disabilities; Unspecified intellectual disabilities	3.6%	1.4%	2
3-digit	296	30 of 37	Bipolar I disorder, single manic episode, unspecified + others	22.4%	8.1%	2.26
3-digit	298	2 of 2	Other and unspecified reactive psychosis; Unspecified psychosis	5.1%	2.3%	1.17
4-digit	2951	2 of 2	Disorganized type schizophrenia, chronic with acute exacerbation; Disorganized type schizophrenia, in remission	1.5%	0.4%	0.83
4-digit	2957	2 of 2	Schizoaffective disorder, unspecified; Schizoaffective disorder, chronic with acute exacerbation	6.6%	1.7%	1.02
leaf	29504	1	Simple type schizophrenia, chronic with acute exacerbation	1.0%	0.2%	2.98
leaf	29524	1	Catatonic type schizophrenia, chronic with acute exacerbation	0.5%	0.3%	1.02
leaf	29530	1	Paranoid type schizophrenia, unspecified	3.6%	0.7%	3.33
leaf	29532	1	Paranoid type schizophrenia, chronic	1.5%	0.6%	2.01
leaf	29534	1	Paranoid type schizophrenia, chronic with acute exacerbation	0.5%	0.3%	0.25
leaf	29535	1	Paranoid type schizophrenia, in remission	1.5%	0.1%	31.94

Level	Node	Codes merged	Description	Cases	Controls	Refit OR
leaf	29560	1	Schizophrenic disorders, residual type, unspecified	2.0%	0.1%	17.29
leaf	29590	1	Unspecified schizophrenia, unspecified	6.1%	1.5%	2.23
leaf	29592	1	Unspecified schizophrenia, chronic	0.5%	0.3%	1.08
leaf	29594	1	Unspecified schizophrenia, chronic with acute exacerbation	1.5%	0.1%	3.55
leaf	29660	1	Bipolar I disorder, most recent episode (or current) mixed, unspecified	1.5%	0.3%	1.5
leaf	29662	1	Bipolar I disorder, most recent episode (or current) mixed, moderate	1.0%	0.2%	3.31
leaf	29663	1	Bipolar I disorder, most recent episode (or current) mixed, severe, without mention of psychotic behavior	1.0%	0.5%	0.15
leaf	29664	1	Bipolar I disorder, most recent episode (or current) mixed, severe, specified as with psychotic behavior	2.6%	0.2%	7.5
leaf	29680	1	Bipolar disorder, unspecified	9.7%	3.2%	1.81
leaf	29681	1	Atypical manic disorder	0.0%	0.3%	0
leaf	29689	1	Other bipolar disorders	2.6%	0.6%	1.79
leaf	2971	1	Delusional disorder	2.0%	0.4%	2.18
leaf	30500	1	Alcohol abuse, unspecified	6.1%	1.1%	3.23
leaf	30501	1	Alcohol abuse, continuous	1.0%	0.2%	0.65
leaf	30503	1	Alcohol abuse, in remission	1.0%	0.2%	1.68
leaf	3051	1	Tobacco use disorder	17.3%	13.8%	0.76
leaf	30520	1	Cannabis abuse, unspecified	0.5%	0.5%	0.51
leaf	30590	1	Other, mixed, or unspecified drug abuse, unspecified	2.6%	0.6%	2.36
leaf	3090	1	Adjustment disorder with depressed mood	0.0%	0.3%	0
leaf	30981	1	Posttraumatic stress disorder	4.6%	0.8%	2.93
leaf	3099	1	Unspecified adjustment reaction	0.0%	0.3%	0

What the inspection shows.

- **It chose different levels of detail for different groups.** Schizophrenia (295.x) and bipolar I mixed and unspecified bipolar codes (296.6x, 296.8x) were mostly kept as **separate codes**. The remaining 30 of 37 codes under 296 were **pooled into one “any other 296” flag**, present in 22.4% of cases versus 8.1% of controls (refit OR 2.26). That is the paper’s central idea: aggregate where codes carry similar, sparse signal, and keep the individual codes where they differ.
- **Some groups were merged wholesale.** All 15 organic-disorder codes (290–294) became a single flag, as did 20 of the 29 codes in section 300–316 (anxiety, personality, substance dependence, depression NOS). The remaining 9 codes in 300–316 (for example alcohol abuse 305.0x, tobacco 305.1, PTSD 309.81) were split out as individual features.
- **A few features are more common in cases.** These include pooled 296 (22.4% vs 8.1%), unspecified schizophrenia 295.90 (6.1% vs 1.5%), alcohol abuse 305.00 (6.1% vs 1.1%), PTSD (4.6% vs 0.8%) and schizoaffective disorder, pooled at 4 digits as 295.7 (6.6% vs 1.7%). These directions are plausible for self-harm risk. Very large odds ratios (for example 31.9 for 295.35) come from codes seen in only a handful of records and should not be read as effect sizes.

5. Methods summary

I applied TSLA (tree-guided rare feature selection and logic aggregation; Chen et al., JASA 2024) to the public CMS DE-SynPUF synthetic Medicare claims, using the paper's nested case-control design for self-harm prediction. Cases were beneficiaries with a first E950-E959 claim and prior history. Each was matched to 10 risk-set controls on sex and birth year, which gave 196 cases and 2,156 records. Features were binary indicators for the 101 ICD-9 mental-disorder codes (290-319) recorded before the index date and present in at least 5 records. The codes were arranged in a five-level hierarchy from full code up to chapter. TSLA expands each tree node into an OR-aggregated indicator and fits a penalized logistic regression. A tuning parameter alpha balances selecting individual codes against aggregating codes that share a parent. The overall penalty strength and alpha were chosen by cross-validated AUC. I compared TSLA with elastic nets on full codes and on 3-digit codes over three 70/30 splits by matched set. On this synthetic data all methods were only slightly better than chance (mean test AUC 0.555-0.594, AUPRC 0.131-0.158 against a random-guess baseline of 0.091), and the differences were within split-to-split noise. The feature inspection showed the intended behavior. TSLA kept clinically distinct schizophrenia and bipolar codes separate while pooling sparse codes in the rest of 296, 290-294 and most of 300-316. Next steps are to pool several DE-SynPUF samples for more cases, reproduce the paper's simulation studies, and use a larger tuning grid.

Software: R, TSLA 0.1.2 (CRAN), glmnet, pROC, PRROC. Seed 2024. Code: 00_config.R through 04_inspect.R.